VE216 Mid-1

Classification of Signals

{continuous ~ {deterministic ~ {energy: $0<E<\infty$
discrete ~ {random ~ {power ~ $0<P<\infty$

even ~ $x_e=\frac{1}{2}(x(t)+x(-t))$ {periodic $x(t)=x(t+T)$
odd ~ $x_o=\frac{1}{2}(x(t)-x(-t))$ {nonperiodic $x[n]=x[n+N]$

conjugate symmetry: $x(-t)=\bar{x}(t)$ $\omega=\frac{2\pi}{T}$ $\Omega=\frac{2\pi}{N}$

Energy & Power

inst. P	$p(t)=\lvert x(t)\rvert^2$	$p[n]=\lvert x[n]\rvert^2$
av. P	$P=\lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}\lvert x(t)\rvert^2dt$	$P=\lim_{N\to\infty}\frac{1}{2N+1}\sum_{n=-N}^{N}\lvert x[n]\rvert^2$
E	$E=\int_{-\infty}^{\infty}\lvert x(t)\rvert^2dt$	$E=\sum_{n=-\infty}^{\infty}\lvert x[n]\rvert^2$

For periodic ~, $P=\frac{1}{T}\int_0^T\lvert x(t)\rvert^2dt$; $P=\frac{1}{N}\sum_{n=0}^{N-1}\lvert x[n]\rvert^2$

Root mean square $X_{rms}=\sqrt{\frac{1}{T_2-T_1}\int_{T_1}^{T_2}\lvert x(t)\rvert^2dt}$

Basic Operations

amplitude scaling $y=cx$ addition $y=x_1+x_2$ multiplication $y=x_1x_2$

differentiation $y(t)=\frac{d}{dt}x(t)$ difference $y[n]=x[n]-x[n-1]$

integration $y(t)=\int_{-\infty}^{t}x(\tau)d\tau$ accumulation $y[n]=\sum_{k=-\infty}^{n}x[k]$

time scaling $y(t)=x(\alpha t)$, $y[n]=x[kn]$. $k\in N^*$ | reflection

time shifting $y(t)=x(t-t_0)$, $y[n]=y[n-n_0]$ | $y(t)=x(-t)$ $y[n]=x[-n]$

Elementary Signals

exponential $x(t)=Be^{\alpha t}$ $x[n]=B\alpha^n$ ← real root

Sinusoidal $x(t)=A\cos(\omega t+\varphi)$ $x[n]=A\cos(\Omega n+\varphi)\Rightarrow \Omega N=2\pi m$ ← imaginary root

exponentially damped sinusoidal $e^{j\theta}=\cos\theta+j\sin\theta$ ← complex root

$x(t)=Ae^{-\alpha t}\sin(\omega t+\varphi)$ $(\alpha>0)$ $x[n]=B\alpha^n\sin(\Omega n+\varphi)$ $(0<|\alpha|<1)$

Let $B=Ae^{j\varphi}$ $A\cos(\omega t+\varphi)=\mathrm{Re}\{Be^{j\omega t}\}$; ← 关系

Step function $u(t)=\begin{cases}1, & t>0\\ 0, & t<0\end{cases}$ $u[n]=\begin{cases}1, & n\geq 0\\ 0, & n<0\end{cases}$

Impulse function {$\delta(t)=0$ for $t=0$, $\int_{-\infty}^{+\infty}\delta(t)dt=1$ $\delta[n]=\begin{cases}1, & n=0\\ 0, & n\neq 0\end{cases}$

$\delta(t)=\frac{d}{dt}u(t)$ $u(t)=\int_{-\infty}^{t}\delta(\tau)d\tau$ $\delta[n]=u[n]-u[n-1]$ $u[n]=\sum_{k=-\infty}^{n}\delta[k]$

* sampling $x(t)\delta(t-t_0)=x(t_0)\delta(t-t_0)$ | time-scaling
* shifting $\int_{-\infty}^{\infty}x(t)\delta(t-t_0)dt=x(t_0)$ | $\delta(at)=\frac{1}{|a|}\delta(t)$
* doublet $\delta^{(1)}(t)=\lim_{\Delta\to 0}\frac{1}{\Delta}(\delta(t+\frac{\Delta}{2})-\delta(t-\frac{\Delta}{2}))$

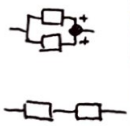
$\int_{-\infty}^{\infty}\delta^{(1)}(t)dt=0$ $\int_{-\infty}^{\infty}f(t)\delta^{(n)}(t-t_0)dt=\frac{d^n}{dt^n}f(t)\big|_{t=t_0}$

Ramp function $r(t)=\begin{cases}t, & t\geq 0\\ 0, & t<0\end{cases}$ $r[n]=\begin{cases}n, & n\geq 0\\ 0, & n<0\end{cases}$

$r(t)=tu(t)$ $u(t)=\frac{d}{dt}r(t)$ $r[n]=nu[n]$ $u[n]=r[n+1]-r[n]$

Interconnection of LTI ~

distributive	$x*h_1+x*h_2=x*(h_1+h_2)$
associative	$(x*h_1)*h_2=x*(h_1*h_2)$
commutative	$h_1*h_2=h_2*h_1$



Convolution Sum

$x[n]=\sum_{k=-\infty}^{\infty}x[k]\delta[n-k]$ $y[n]=\sum_{k=-\infty}^{\infty}x[k]h[n-k]=x[n]*h[n]$

法一 ① 用δ表示x ② 用h表示y

法二 ① 令x=δ求出h ② reflect & shift 确定范围 $x[k]$ $h[n-k]$

③ 求 $W_n[k]=x[k]h[n-k]$ ④ $y[n]=\sum_{k=-\infty}^{\infty}W_n[k]$ * k换为 n-k

Convolution Integral

$x(t)=\int_{-\infty}^{\infty}x(\tau)\delta(t-\tau)d\tau$ $y(t)=\int_{-\infty}^{\infty}x(\tau)h(t-\tau)d\tau=x(t)*h(t)$

① 令 x=δ求出h ② reflect & shift 确定范围 $x(\tau)$ $h(t-\tau)$

③ 求 $W_t(\tau)=x(\tau)h(t-\tau)$ ④ $y(t)=\int_{t=-\infty}^{\infty}W_t(\tau)d\tau$

Properties of Systems (LTI)

memoryless - depend only on current input $h(\tau)=C\delta(\tau)$ $h[k]=C\delta[k]$

causal - depend only on past/present input $h(\tau)=0$ for $\tau<0$ $h[k]=0$ for $k<0$

stable - BI⇒BO $\sum_{k=-\infty}^{\infty}\lvert h[k]\rvert<\infty$ $\int_{-\infty}^{\infty}\lvert h(\tau)\rvert d\tau<\infty$

invertible $h(t)*h^{inv}(t)=\delta(t)$ $h[n]*h^{inv}[n]=\delta[n]$

* ① 令 x=δ 得 y=h ② 令 y=δ 得 x=h^{inv}

Step Response $s(t)=\int_{-\infty}^{t}h(\tau)d\tau$ $s[n]=\sum_{k=-\infty}^{n}h[k]$

$h(t)=\frac{d}{dt}s(t)$ $h[n]=s[n]-s[n-1]$

Differential Equation

$\sum_{k=0}^{N}a_k\frac{d^k}{dt^k}y(t)=\sum_{k=0}^{M}b_k\frac{d^k}{dt^k}x(t)$

① $\sum_{k=0}^{N}a_kr^k=0\Rightarrow y^{(h)}(t)=\sum_{i=1}^{N}C_ie^{r_it}$ 若r_j为p重根 $e^{r_jt}, te^{r_jt}, \dots, t^{p-1}e^{r_jt}$

② 设$y^{(p)}(t)$. 求 ③ $y(t)=y^{(h)}(t)+y^{(p)}(t)$ 代入初始值求.

初始值: $y(t)\big|_{t=0^-}$ $\frac{d}{dt}y(t)\big|_{t=0^-}$ … $\frac{d^{N-1}}{dt^{N-1}}y(t)\big|_{t=0^-}\Rightarrow t=0^+$

Difference Equation

$\sum_{k=0}^{N}a_ky[n-k]=\sum_{k=0}^{M}b_kx[n-k]$

① $\sum_{k=0}^{N}a_kr^{N-k}=0\Rightarrow y^{(h)}[n]=\sum_{i=1}^{N}C_ir_i^n$ 若r_j为p重根 $r_j^n, nr_j^n \dots n^{p-1}r_j^n$

② 设$y^{(p)}[n]$ 求 ③ $y[n]=y^{(h)}[n]+y^{(p)}[n]$ 代入初始值. 求

初始值: $y[-N], y[-N+1] \dots y[-1]\Rightarrow y[0]\, y[1] \dots y[N-1]$

$x(t)$	$y^{(p)}(t)$	$x[n]$	$y^{(p)}[n]$
1	C	1	C
t	C_1t+C_2	n	C_1n+C_2
$e^{-\alpha t}$	$Ce^{-\alpha t}$	α^n	$C\alpha^n$
$\cos(\omega t+\varphi)$	$C_1\cos(\omega t+\varphi)+C_2\sin(\omega t+\varphi)$	"	

connection $\frac{d}{dt}x(t)=\frac{x[n]-x[n-1]}{T_s}$

* nautral response $y^{(n)}(t)[n]\Leftarrow y^{(h)}(0)$ / $y^{(h)}[-1]$
* forced response $y^{(f)}(t)[n]\Leftarrow$ 令初始值为0

Fourier

$x[n]=e^{j\Omega n}$ $y[n]=H(e^{j\Omega})e^{j\Omega n}$ $(H(e^{j\Omega})=\sum_{k=-\infty}^{\infty}h[k]e^{-j\Omega k})$

$x(t)=e^{j\omega t}$ $y(t)=H(j\omega)e^{j\omega t}$ $(H(j\omega)=\int_{-\infty}^{\infty}h(\tau)e^{-j\omega\tau}d\tau)$

For DTFS $\hat{x}[n]=\sum_{k=0}^{N-1}A[k]e^{jk\Omega_0 n}$ $MSE=\frac{1}{N}\sum_{n=0}^{N-1}\lvert x[n]-\hat{x}[n]\rvert^2$ mean-square error

For FS $\hat{x}(t)=\sum_{k=-\infty}^{\infty}A[k]e^{jk\omega_0 t}$ $MSE=\frac{1}{T}\int_0^T\lvert x(t)-\hat{x}(t)\rvert^2dt$

function of frequency

	periodic	nonperiodic
continuous	FS	FT
discrete	DTFS	DTFT

$C=a+jb$

$C=|C|e^{j\arg\{C\}}$

where $|C|=\sqrt{a^2+b^2}$ $\arg\{C\}=\arctan(\frac{b}{a})$

$\cos^2\theta=\frac{1+\cos 2\theta}{2}$ $\sin^2\theta=\frac{1-\cos 2\theta}{2}$

2015.3.18

VE216 Mid-2

Dirichlet Conditions

For FS ① bounded

② finite max/min in one period

③ finite discontinuities in one period

For FT ① absolutely integrable $\int_{-\infty}^{\infty}|x(t)|dt<\infty$

② finite max/min/discontinuities in any finite interval

③ size of discontinuities is finite

* $\hat{x}(t)$ converges to the midpoint if dis~

Trigonometric FS

$x(t)=B[0]+\sum_{k=1}^{\infty}B[k]\cos(k\omega_0 t)+A[k]\sin(k\omega_0 t)$

$$\begin{cases} B[0]=\frac{1}{T}\int_0^T x(t)dt \\ B[k]=\frac{2}{T}\int_0^T x(t)\cos(k\omega_0 t)dt \\ A[k]=\frac{2}{T}\int_0^T x(t)\sin(k\omega_0 t)dt \end{cases} \qquad \begin{cases} B[k]=X[k]+X[-k] \\ A[k]=j(X[k]-X[-k]) \end{cases}$$

Properties of FT (略)

Time reversal $x(-t)\xleftrightarrow{FT}X(-j\omega)$

Conjugation $x^*(t)\xleftrightarrow{FT}X^*(-j\omega)$

Even-odd decomp $\begin{cases} x_e(t)\xleftrightarrow{FT}Re\{X(j\omega)\} \\ x_o(t)\xleftrightarrow{FT}Im\{X(j\omega)\} \end{cases}$

How to 求 System Frequency Response

[I] $y(t)=h(t)*x(t)\xleftrightarrow{FT}Y(j\omega)=H(j\omega)X(j\omega)$

[II] $\sum_{k=0}^{N}a_k\frac{d^k}{dt^k}y(t)=\sum_{k=0}^{M}b_k\frac{d^k}{dt^k}x(t)$

$\xleftrightarrow{FT}\sum_{k=0}^{N}a_k(j\omega)^kY(j\omega)=\sum_{k=0}^{M}b_k(j\omega)^kX(j\omega)$

$$\Rightarrow H(j\omega)=\frac{Y(j\omega)}{X(j\omega)}=\frac{\sum_{k=0}^{M}b_k(j\omega)^k}{\sum_{k=0}^{N}a_k(j\omega)^k}=\frac{c_1}{j\omega-d_1}+\cdots+\frac{c_N}{j\omega-d_N}$$

$$=\sum_{k=0}^{N}e^{d_k t}(d_k<0)$$

special cases $\begin{cases} ① \deg Y>\deg X. \\ ② \text{complex roots} \\ ③ \text{multiple roots} \end{cases}$

法一: e.g. $X(j\omega)=\frac{A_1}{(j\omega+1)^2}+\frac{A_2}{j\omega+1}$

$A_1=(j\omega+1)^2X(j\omega)|_{j\omega=-1}$

$A_2=\frac{d}{d(j\omega)}\{(j\omega+1)^2X(j\omega)\}|_{j\omega=-1}$

法二: 令 $j\omega=0$ 求解.

Time-bandwidth Property

Duration $T_d=\sqrt{\frac{\int_{-\infty}^{\infty}t^2|x(t)|^2dt}{\int_{-\infty}^{\infty}|x(t)|^2dt}}$ time-domain; for slower response

Bandwidth $B_\omega=\sqrt{\frac{\int_{-\infty}^{\infty}\omega^2|X(j\omega)|^2d\omega}{\int_{-\infty}^{\infty}|X(j\omega)|^2d\omega}}$ frequency-domain; for wider bandpass

Uncertainty principle $T_dB_\omega\geq\frac{1}{2}$ or $\sigma_t^2\sigma_\omega^2\geq\frac{1}{4}$

$\int_{-\infty}^{\infty}e^{-x^2/2\sigma^2}dx=\sigma\sqrt{2\pi}.\ \sigma>0$ $\int_{-\infty}^{\infty}x^2e^{-x^2/2\sigma^2}dx=\sigma^3\sqrt{2\pi}.\ \sigma>0$	$\mathrm{sinc}\,x=\frac{\sin(\pi x)}{\pi x}$

Sampling

$x_s(t)=\sum_{n=-\infty}^{\infty}x[n]\delta(t-nT_s)=\sum_{n=-\infty}^{\infty}x(nT_s)\delta(t-nT_s)$

$=x(t)p(t)$ (impulse train)

FT: $X_s(j\omega)=\frac{1}{T_s}\sum_{k=-\infty}^{\infty}X(j(\omega-k\omega_s))$

$x[n]\xleftrightarrow{DTFS}X(e^{j\omega})=X_s(j\omega)|_{\omega=\frac{\Omega}{T_s}}$

Nyquist rate

Shannon Sampling Theorem: $\omega_s>2W\Rightarrow$ no aliasing

Ideal Reconstruction

$X(j\omega)=X_s(j\omega)H_r(j\omega)$ where $H_r(j\omega)=\begin{cases} T_s, & |\omega|\leq\omega_s/2 \\ 0, & |\omega|>\omega_s/2 \end{cases}$

$x(t)=x_s(t)*h_r(t)$, where $h_r(t)=\frac{T_s\sin(\frac{\omega_s}{2}t)}{\pi t}$

$=\sum_{n=-\infty}^{\infty}x[n]\sin(\omega_s(t-nT_s)/2\pi)$

Partial Reconstruction (Zero-order Holder)

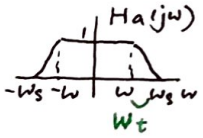
$h_0(t)=\begin{cases} 1, & 0<t<T_s \\ 0, & t<0, t>T_s \end{cases}\xleftrightarrow{FT}H_0(j\omega)=2e^{-j\omega T_s/2}\frac{\sin(\omega T_s/2)}{\omega}$

$H_c(j\omega)=\begin{cases} \frac{\omega T_s}{2\sin(\omega T_s/2)}, & |\omega|<\omega_m \\ 0, & |\omega|>\omega_s-\omega_m \end{cases}$

Oversampling

$\omega_s-W_s>W$

$(W_s=W_t+W)$ $W_t<\omega_s-2W$



Discrete-time Processing

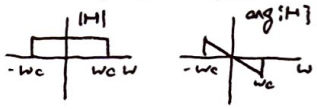
$x(t)\to\boxed{H_a(j\omega)}\xrightarrow{x_a(t)}\boxed{\text{Sample}}\xrightarrow{x[n]}\boxed{H}\xrightarrow{y[n]}\boxed{H_0(j\omega)}\xrightarrow{y_0(t)}\boxed{H_c(j\omega)}\to y(t)$

$G(j\omega)=[\frac{1}{T_s}H_0(j\omega)H_c(j\omega)H_a(j\omega)]H(e^{j\omega T_s})$

Ideal Low-pass Filter

$H(j\omega)=\begin{cases} e^{-j\omega t_0} & |\omega|\leq\omega_c \\ 0 & |\omega|>\omega_c \end{cases}\Rightarrow h(t)=\frac{\sin(\omega_c(t-t_0))}{\pi(t-t_0)}$

$=\frac{\omega_c}{\pi}\mathrm{sinc}(\frac{\omega_c}{\pi}(t-t_0))$



Proof of FT

$\int_{-\infty}^{\infty}x(t)e^{-j\omega t}dt=\int_{-\infty}^{\infty}[\frac{1}{2\pi}\int_{-\infty}^{\infty}X(jr)e^{jrt}dr]\cdot e^{-j\omega t}dt$

$=\int_{-\infty}^{\infty}[\frac{1}{2\pi}X(jr)\int_{-\infty}^{\infty}e^{j(r-\omega)t}dt]dr$

$=\int_{-\infty}^{\infty}\frac{1}{2\pi}X(jr)2\pi\delta(r-\omega)dr=X(j\omega)$

$\int_{-\infty}^{\infty}e^{-j\omega t}dt=\int_{-\infty}^{0}e^{-j\omega t}dt+\int_{0}^{\infty}e^{-j\omega t}dt$

$=\lim_{a\to0^+}(\int_{-\infty}^{0}e^{(a-j\omega t)}dt+\int_{0}^{\infty}e^{(-a-j\omega t)}dt)$

$=\lim_{a\to0^+}(\frac{1}{a-j\omega}+\frac{1}{a+j\omega})=\lim_{a\to0^+}\frac{2a}{a^2+\omega^2}=\begin{cases} \infty & \omega=0 \\ 0 & \omega\neq0 \end{cases}$

$\int_{-\infty}^{\infty}\frac{2a}{a^2+\omega^2}d\omega=\cdots=2\pi$

$\Rightarrow\int_{-\infty}^{\infty}e^{-j\omega t}dt=2\pi\delta(\omega)$

2015.4.3

VE216 Mid-2

Figure 1: FS

Time Domain	*Frequency Domain*
$x(t) = \sum_{k=-\infty}^{\infty} X[k]e^{jk\omega_o t}$ *Period* $= T$	$X[k] = \frac{1}{T}\int_0^T x(t)e^{-jk\omega_o t}\,dt$ $\omega_o = \frac{2\pi}{T}$
$x(t) = \begin{cases} 1, & \lvert t\rvert \le T_o \\ 0, & T_o < \lvert t\rvert \le T/2 \end{cases}$	$X[k] = \frac{\sin(k\omega_o T_o)}{k\pi}$
$x(t) = e^{jp\omega_o t}$	$X[k] = \delta[k-p]$
$x(t) = \cos(p\omega_o t)$	$X[k] = \frac{1}{2}\delta[k-p] + \frac{1}{2}\delta[k+p]$
$x(t) = \sin(p\omega_o t)$	$X[k] = \frac{1}{2j}\delta[k-p] - \frac{1}{2j}\delta[k+p]$
$x(t) = \sum_{p=-\infty}^{\infty}\delta(t-pT)$	$X[k] = \frac{1}{T}$

Figure 2: FT

Time Domain	*Frequency Domain*
$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)e^{j\omega t}\,d\omega$	$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\,dt$
$x(t) = \begin{cases} 1, & \lvert t\rvert \le T_o \\ 0, & \text{otherwise} \end{cases}$	$X(j\omega) = \frac{2\sin(\omega T_o)}{\omega}$
$x(t) = \frac{1}{\pi t}\sin(Wt)$	$X(j\omega) = \begin{cases} 1, & \lvert\omega\rvert \le W \\ 0, & \text{otherwise} \end{cases}$
$x(t) = \delta(t)$	$X(j\omega) = 1$
$x(t) = 1$	$X(j\omega) = 2\pi\delta(\omega)$
$x(t) = u(t)$	$X(j\omega) = \frac{1}{j\omega} + \pi\delta(\omega)$
$x(t) = e^{-at}u(t), \quad \mathrm{Re}\{a\} > 0$	$X(j\omega) = \frac{1}{a+j\omega}$
$x(t) = te^{-at}u(t), \quad \mathrm{Re}\{a\} > 0$	$X(j\omega) = \frac{1}{(a+j\omega)^2}$
$x(t) = e^{-a\lvert t\rvert}, \quad a > 0$	$X(j\omega) = \frac{2a}{a^2+\omega^2}$
$x(t) = \frac{1}{\sqrt{2\pi}}e^{-t^2/2}$	$X(j\omega) = e^{-\omega^2/2}$

Figure 3: FT of Periodic

Periodic Time-Domain Signal	*Fourier Transform*
$x(t) = \sum_{k=-\infty}^{\infty} X[k]e^{jk\omega_o t}$	$X(j\omega) = 2\pi\sum_{k=-\infty}^{\infty} X[k]\delta(\omega - k\omega_o)$
$x(t) = \cos(\omega_o t)$	$X(j\omega) = \pi\delta(\omega-\omega_o) + \pi\delta(\omega+\omega_o)$
$x(t) = \sin(\omega_o t)$	$X(j\omega) = \frac{\pi}{j}\delta(\omega-\omega_o) - \frac{\pi}{j}\delta(\omega+\omega_o)$
$x(t) = e^{j\omega_o t}$	$X(j\omega) = 2\pi\delta(\omega-\omega_o)$
$x(t) = \sum_{n=-\infty}^{\infty}\delta(t-nT_s)$	$X(j\omega) = \frac{2\pi}{T_s}\sum_{k=-\infty}^{\infty}\delta\left(\omega - k\frac{2\pi}{T_s}\right)$
$x(t) = \begin{cases} 1, & \lvert t\rvert \le T_o \\ 0, & T_o < \lvert t\rvert < T/2 \end{cases}$ $x(t+T) = x(t)$	$X(j\omega) = \sum_{k=-\infty}^{\infty}\frac{2\sin(k\omega_o T_o)}{k}\delta(\omega - k\omega_o)$

Figure 4: Properties of FT

Property	*Fourier Transform* $x(t) \xleftrightarrow{FT} X(j\omega)$ $y(t) \xleftrightarrow{FT} Y(j\omega)$
Linearity	$ax(t) + by(t) \xleftrightarrow{FT} aX(j\omega) + bY(j\omega)$
Time shift	$x(t-t_o) \xleftrightarrow{FT} e^{-j\omega t_o}X(j\omega)$
Frequency shift	$e^{j\gamma t}x(t) \xleftrightarrow{FT} X(j(\omega-\gamma))$
Scaling	$x(at) \xleftrightarrow{FT} \frac{1}{\lvert a\rvert}X\left(\frac{j\omega}{a}\right)$
Differentiation in time	$\frac{d}{dt}x(t) \xleftrightarrow{FT} j\omega X(j\omega)$
Differentiation in frequency	$-jtx(t) \xleftrightarrow{FT} \frac{d}{d\omega}X(j\omega)$
Integration/ Summation	$\int_{-\infty}^{t} x(\tau)\,d\tau \xleftrightarrow{FT} \frac{X(j\omega)}{j\omega} + \pi X(j0)\delta(\omega)$
Convolution	$\int_{-\infty}^{\infty} x(\tau)y(t-\tau)\,d\tau \xleftrightarrow{FT} X(j\omega)Y(j\omega)$
Multiplication	$x(t)y(t) \xleftrightarrow{FT} \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\nu)Y(j(\omega-\nu))\,d\nu$
Parseval's Theorem	$\int_{-\infty}^{\infty}\lvert x(t)\rvert^2\,dt = \frac{1}{2\pi}\int_{-\infty}^{\infty}\lvert X(j\omega)\rvert^2\,d\omega$
Duality	$X(jt) \xleftrightarrow{FT} 2\pi x(-\omega)$
Symmetry	$x(t)$ real $\xleftrightarrow{FT} X^*(j\omega) = X(-j\omega)$ $x(t)$ imaginary $\xleftrightarrow{FT} X^*(j\omega) = -X(-j\omega)$ $x(t)$ real and even $\xleftrightarrow{FT} \mathrm{Im}\{X(j\omega)\} = 0$ $x(t)$ real and odd $\xleftrightarrow{FT} \mathrm{Re}\{X(j\omega)\} = 0$

Figure 5: Properties of FS

Fourier Series $x(t) \xleftrightarrow{FS;\omega_o} X[k]$ $y(t) \xleftrightarrow{FS;\omega_o} Y[k]$ *Period* $= T$	—
$ax(t) + by(t) \xleftrightarrow{FS;\omega_o} aX[k] + bY[k]$	—
$x(t-t_o) \xleftrightarrow{FS;\omega_o} e^{-jk\omega_o t_o}X[k]$	$\int_0^T x(\tau)y(t-\tau)\,d\tau \xleftrightarrow{FS;\omega_o} TX[k]Y[k]$
$e^{jk_o\omega_o t}x(t) \xleftrightarrow{FS;\omega_o} X[k-k_o]$	$x(t)y(t) \xleftrightarrow{FS;\omega_o} \sum_{l=-\infty}^{\infty} X[l]Y[k-l]$
$x(at) \xleftrightarrow{FS;a\omega_o} X[k]$	$\frac{1}{T}\int_0^T \lvert x(t)\rvert^2\,dt = \sum_{k=-\infty}^{\infty}\lvert X[k]\rvert^2$
$\frac{d}{dt}x(t) \xleftrightarrow{FS;\omega_o} jk\omega_o X[k]$	$x[n] \xleftrightarrow{DTFT} X(e^{j\Omega})$ $X(e^{jt}) \xleftrightarrow{FS;1} x[-k]$

2015.4.3

Laplace Transform

$\mathcal{L}\{x(t)\} = FT\{x(t)e^{-\sigma t}\}$; ROC: $\int_{-\infty}^{\infty}|x(t)e^{-\sigma t}|dt<\infty$, $x(t) \xleftrightarrow{\mathcal{L}u} X(s)=\int_{0^-}^{\infty}x(t)e^{-st}dt$

$x(t)=\frac{1}{2\pi j}\int_{\sigma-j\infty}^{\sigma+j\infty}X(s)e^{st}dt$	$X(s)=\int_{-\infty}^{\infty}x(t)e^{-st}dt$	ROC
$u(t)$	$\frac{1}{s}$	$Re(s)>0$
$tu(t)$	$\frac{1}{s^2}$	$Re(s)>0$
$\delta(t-\tau),\ \tau\geq 0$	$e^{-s\tau}$	all s
$e^{-at}u(t)$	$\frac{1}{s+a}$	$Re(s)>-a$
$te^{-at}u(t)$	$\frac{1}{(s+a)^2}$	$Re(s)>-a$
$[\cos(\omega_1 t)]u(t)$	$\frac{s}{s^2+\omega_1^2}$	$Re(s)>0$
$[\sin(\omega_1 t)]u(t)$	$\frac{\omega_1}{s^2+\omega_1^2}$	$Re(s)>0$
$[e^{-at}\cos(\omega_1 t)]u(t)$	$\frac{s+a}{(s+a)^2+\omega_1^2}$	$Re(s)>-a$
$[e^{-at}\sin(\omega_1 t)]u(t)$	$\frac{\omega_1}{(s+a)^2+\omega_1^2}$	$Re(s)>-a$

$ax(t)+by(t)$	$aX(s)+bY(s)$	$aX(s)+bY(s)$	At least $R_x\cap R_y$
$x(t-\tau)$	$e^{-s\tau}X(s)$ if $x(t-\tau)u(t)=x(t-\tau)u(t-\tau)$	$e^{-s\tau}X(s)$	R_x
$e^{s_0 t}x(t)$	$X(s-s_0)$	$X(s-s_0)$	$R_x+Re(s_0)$
$x(at)$	$\frac{1}{a}X(\frac{s}{a}),a>0$	$\frac{1}{\lvert a\rvert}X(\frac{s}{a})$	$\frac{R_x}{\lvert a\rvert}$
$x(t)*y(t)$	$X(s)Y(s)$ if $x(t)=y(t)=0$ for $t<0$	$X(s)Y(s)$	At least $R_x\cap R_y$
$-tx(t)$			
$\frac{d}{dt}x(t)$	$sX(s)-x(0^-)$	$sX(s)$	At least R_x
$\int_{-\infty}^{t}x(\tau)d\tau$	$\frac{1}{s}\int_{-\infty}^{0^-}x(\tau)d\tau+\frac{X(s)}{s}$	$\frac{X(s)}{s}$	At least $R_x\cap\{Re(s)>0\}$

$\delta(t-\tau),\ \tau<0$	$e^{-s\tau}$	all s
$-u(-t)$	$\frac{1}{s}$	$Re(s)<0$
$-tu(-t)$	$\frac{1}{s^2}$	$Re(s)<0$
$-e^{-at}u(-t)$	$\frac{1}{s+a}$	$Re(s)<-a$
$-te^{-at}u(-t)$	$\frac{1}{(s+a)^2}$	$Re(s)<-a$

$X(s)=\frac{b_m\prod_{k=1}^{M}(s-c_k)}{\prod_{k=1}^{N}(s-d_k)}$ zeros ∘ poles ×

property of ROC: ① no × ② ||| ③ if $x(t)$ finite duration & ab. integrable ⇒ ROC: s

④ right-sided ⇒ $\sigma>\sigma_p$
left-sided ⇒ $\sigma<\sigma_n$
two-sided ⇒ $\sigma_p<\sigma<\sigma_n$

z-Transform

$z\{x(t)\} = DTFT\{x[n]r^{-n}\}$; ROC: $\sum_{n=-\infty}^{\infty}|x[n]|r^{-n}<\infty$

$\frac{d^n}{dt^n}x(t) \xleftrightarrow{\mathcal{L}u} s^nX(s)-\frac{d^{n-1}}{dt^{n-1}}x(t)\Big|_{t=0^-}-\cdots-s^{n-1}x(0^-)$

$x[n]=\frac{1}{2\pi j}\oint X(z)z^{n-1}dz$	$X[z]=\sum_{n=-\infty}^{\infty}x[n]z^{-n}$	ROC
$\delta[n]$	1	all z
$u[n]$	$\frac{1}{1-z^{-1}}$	$\lvert z\rvert>1$
$\alpha^n u[n]$	$\frac{1}{1-\alpha z^{-1}}$	$\lvert z\rvert>\lvert\alpha\rvert$
$n\alpha^n u[n]$	$\frac{\alpha z^{-1}}{(1-\alpha z^{-1})^2}$	$\lvert z\rvert>\lvert\alpha\rvert$
$[\cos(\Omega_1 n)]u[n]$	$\frac{1-z^{-1}\cos\Omega_1}{1-z^{-1}2\cos\Omega_1+z^{-2}}$	$\lvert z\rvert>1$
$[\sin(\Omega_1 n)]u[n]$	$\frac{z^{-1}\sin\Omega_1}{1-z^{-1}2\cos\Omega_1+z^{-2}}$	$\lvert z\rvert>1$
$[r^n\cos(\Omega_1 n)]u[n]$	$\frac{1-z^{-1}r\cos\Omega_1}{1-z^{-1}2r\cos\Omega_1+r^2z^{-2}}$	$\lvert z\rvert>r$
$[r^n\sin(\Omega_1 n)]u[n]$	$\frac{z^{-1}r\sin\Omega_1}{1-z^{-1}2r\cos\Omega_1+r^2z^{-2}}$	$\lvert z\rvert>r$

$ax[n]+by[n]$	$aX(z)+bY(z)$	$aX(z)+bY(z)$	At least $R_x\cap R_y$
$x[n-k]$	see below	$z^{-k}X(z)$	R_x, except $\lvert z\rvert=0,\infty$
$\alpha^n x[n]$	$X(\frac{z}{\alpha})$	$X(\frac{z}{\alpha})$	$\lvert\alpha\rvert R_x$
$x[-n]$	—	$X(\frac{1}{z})$	$\frac{1}{R_x}$
$x[n]*y[n]$	$X(z)Y(z)$ if $x[n]=y[n]=0$ for $n<0$	$X(z)Y(z)$	At least $R_x\cap R_y$
$nx[n]$	$-z\frac{d}{dz}X(z)$	$-z\frac{d}{dz}X(z)$	R_x, except $z=0$

$u[-n-1]$	$\frac{1}{1-z^{-1}}$	$\lvert z\rvert<1$
$-\alpha^n u[-n-1]$	$\frac{1}{1-\alpha z^{-1}}$	$\lvert z\rvert<\lvert\alpha\rvert$
$-n\alpha^n u[-n-1]$	$\frac{\alpha z^{-1}}{(1-\alpha z^{-1})^2}$	$\lvert z\rvert<\lvert\alpha\rvert$

$X(z)=\frac{b_0\prod_{k=1}^{M}(1-c_kz^{-1})}{\prod_{k=1}^{N}(1-d_kz^{-1})}$ zeros ∘ poles ×

property of ROC: ① no × ② ⓪ ③ if $x[n]$ finite duration ⇒ ROC: z only

④ right-sided ⇒ $r>r_+$
left-sided ⇒ $r<r_-$
two-sided ⇒ $r_+<r<r_-$

$x[n-k] \xleftrightarrow{z_u} x[-k]+x[-k+1]z^{-1}+\cdots+x[-1]z^{-k+1}+z^{-k}X(z)$

$x[n+k] \xleftrightarrow{z_u} -x[0]z^k-x[1]z^{k-1}-\cdots-x[k-1]z+z^kX(z)\ (k>0)$

$\delta[n],\ X(z)=1$

$\delta[n+1]\ X(z)=z^{-1}$

$\delta[n-1]\ X(z)=z$

	Initial Value Th.	Final Value Th.
n_y	$\lim_{s\to\infty}sX(s)=x(0^+)$	$\lim_{s\to 0}sX(s)=\lim_{t\to\infty}x(t)$
n_z	$\lim_{z\to\infty}X(z)=x[0]$ if $x[n]=0$ for $n<0$	

⇒ # of ∘ < # of × for causal $x[n]$

LT for circuit analysis

$V_R(s)=RI_R(s)$

$V_L(s)=sLI_L(s)-Li_L(0^-)$

$I_C(s)=sCV_C(s)-Cv_C(0^-)$

About DTFS

$X(e^{j\omega})=\sum_{n=-\infty}^{\infty}x[n]e^{-j\omega n}$

$\alpha^n u[n],\ \lvert\alpha\rvert<1 \leftrightarrow \frac{1}{1-\alpha e^{-j\omega}}$

$\begin{cases}1, & \lvert n\rvert<M\\ 0, & \lvert n\rvert>M\end{cases} \leftrightarrow \frac{\sin(\Omega(2M+1)/2)}{\sin(\Omega/2)}$

$\frac{\sin(Wn)}{\pi n} \leftrightarrow \begin{cases}1, & \lvert\Omega\rvert<W\\ 0, & W<\lvert\Omega\rvert<\pi\end{cases}$

$\delta[n] \leftrightarrow 1$

$1 \leftrightarrow 2\pi\delta(\Omega)$

$\lvert\alpha\rvert<1,\ \lvert\alpha\rvert<1 \leftrightarrow \frac{1-\alpha^2}{1-2\alpha\cos\Omega+\alpha^2}$

phase response

$\arg\{H(j\omega)\}=\arg\{b_m\}+\sum\arg\{j\omega-z\}-\sum\arg\{j\omega-d\}$

$A_k(d_k)^n u[n] \xleftrightarrow{z} \frac{A_k}{1-d_kz^{-1}}$ ROC: $\lvert z\rvert>d_k$

$A_{im}\frac{(n+1)\cdots(n+m-1)}{(m-1)!}(d_k)^n u[n] \xleftrightarrow{z} \frac{A_{im}}{(1-d_kz^{-1})^m}$

For ②, causal ⇒ right-sided

stable ⇒ ROC ⊃ unit circle

For ①, causal ⇒ $h(t)=0$ for $t<0$ ⇒ right-sided

stable ⇒ $\int_{-\infty}^{\infty}|x(t)e^{-\sigma t}|dt<\infty$. ROC ⊃ $Re(s)=0$

causal & stable ⇒ poles in unit circle

causal & stable ⇒ poles in l.h.

inverse $H^{inv}=\frac{1}{H}$. zeros.

zeros in l.h / unit circle

minimum phase system

$x(t)\overset{FT}{=}\frac{1}{2\pi}\int_{-\infty}^{\infty}X(j\omega)e^{j\omega t}dt,\ X(j\omega)=\int_{-\infty}^{\infty}x(t)e^{-j\omega t}dt$

DTFS: $x[n]=\sum_{k=0}^{N-1}X[k]e^{jk\omega_0 n},\ X[k]=\frac{1}{N}\sum_{n=0}^{N-1}x[n]e^{-jk\omega_0 n}$

DTFT: $x[n]=\frac{1}{2\pi}\int_{-\pi}^{\pi}X(e^{j\omega})e^{j\omega n}d\omega;\ X(e^{j\omega})=\sum_{n=-\infty}^{\infty}x[n]e^{-j\omega n}$

$(\omega_{k+1}-\omega_k)S[k]=X[k]$

Let $G_m>1$. ⇒ Tx for.

$T(s)=\frac{Y(s)}{G(s)}=\frac{GH}{1+GH}$

closed-loop ~ if $H=0$ Tx for. if $H=$ const. $y_d=1$

$E(s)=Y_d(s)-T(s)$ let $G_m=1$, $Y=Y_d+N$

Control System

open-loop ~ $y_d \to H \to G \to y$

Stability

First-order $T(s)=\frac{B(s)}{A(s)}=\frac{b_0}{s+a_0}$

Second-order $T(s)=\frac{B(s)}{A(s)}=\frac{b_1s+b_0}{s^2+a_1s+a_0}$

⇒ stable if coefficients of $A(s)$ are positive

Higher-order $A(s)=\prod_k(s+c_k)\prod(s^2+d_is+d_0)$

⇒ not stable if coefficients are not all positive

Routh-Hurwitz Criterion

$b_i=-\frac{\begin{vmatrix}a_n & a_{n-2i}\\ a_{n-1} & a_{n-2i-1}\end{vmatrix}}{a_{n-1}}$

$c_i=-\frac{\begin{vmatrix}b_1 & b_{i+1}\\ a_{n-1} & a_{n-2i-1}\end{vmatrix}}{b_1}$

s^n	a_n	$a_{n-2}\cdots$
s^{n-1}	a_{n-1}	$a_{n-3}\cdots$
s^{n-2}	b_1	$b_2\cdots$
s^{n-3}	c_1	$c_2\cdots$
$\vdots$	$\vdots$	$\vdots$

⊛ roots of sign change = # of roots in r.h. ⊛ row of zero ⇒ Usage of stability ⊛ one zero ⇒ $A(s)$ in l.h. ⊛ # of sign change

Block Diagram

Cascade $H=\prod_{i=1}^{p}H_i$: $-H_1-H_2-\cdots-H_p-$

parallel $H=\sum_{i=1}^{p}H_i$: $-\boxed{H_1}-$ $\boxed{H_2}$

$H(z)=\frac{b_0+b_1z^{-1}+b_2z^{-2}}{1+a_1z^{-1}+a_0z^{-2}}$

$H(s)=\frac{b_2+b_1s^{-1}+b_0}{1+a_1s^{-1}+a_0}$

$X(z)$ $Y(z)$ b_0 b_1 b_2 $-a_1$ $-a_0$ z^{-1}

$X(s)$ $Y(s)$ b_2 b_1 b_0 $-a_1$ $-a_0$ $\frac{1}{s}$

feedback

input $x(t)$, error $e(t)$, output $y(t)$, Gain, feedback $h(t)$

★ compare negative adder positive

$T(s)=\frac{Y(s)}{X(s)}=\frac{G(s)}{1+G(s)H(s)}$

$S_G^T=\frac{dT/T}{dG/G}=\frac{1}{1+GH}=\frac{1}{F}$

$X(s)$ $E(s)$ $G(s)$ $Y(s)$ $H(s)$ $R(s)$

y_d y H G $N(s)$ $y=Gx+n$

effect on noise

without feedback $y=Gx+n$

with feedback $y=\frac{G}{1+GH}x+\frac{1}{H(1+GH)}n$

distortion analysis

with feedback $e=a_1y+a_2y^2 \Rightarrow \frac{dx}{dy}=(a_1+H)(1+\frac{2a_2}{a_1+H}y)$

without feedback $x=a_1y+a_2y^2 \Rightarrow \frac{dx}{dy}=a_1(1+\frac{2a_2}{a_1}y)$

Reduced distortion $\Delta=\frac{1}{F}$

nonlinear $x=a_1y$ real model $x=a_1y+a_2y^2$